QUANTITATIVE RESEARCH BRIEF

Football Betting Arbitrage & Forecasting

Offensive/Defensive Points per Play

A probability-driven framework for winner, point spread, and game-total (Over/Under) markets in NFL and college football.

Central distinction. A model that predicts outcomes more accurately than a sportsbook may uncover *positive-expected-value* opportunities. It does not create risk-free arbitrage. A genuine cross-book arbitrage requires compatible prices that cover every possible settlement outcome at a positive net payoff. Execution and settlement risks still matter.

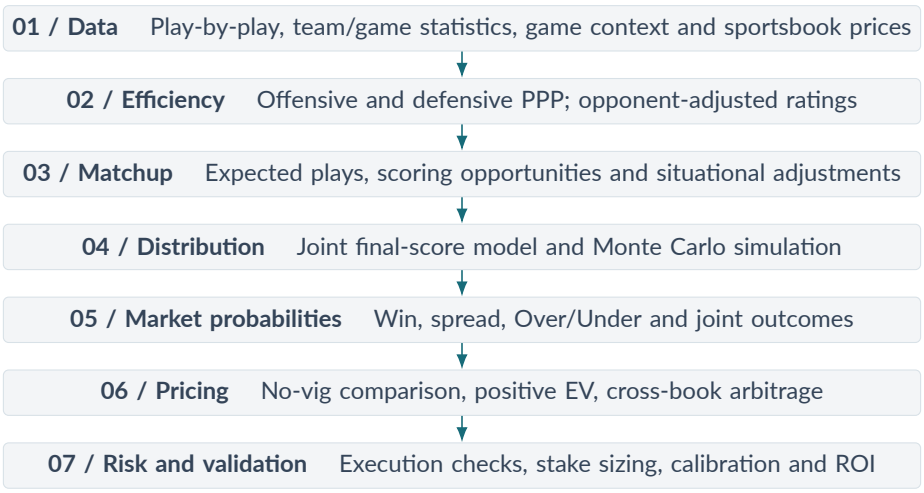
Forecasting engine
PPP and opponent quality
Expected plays and game pace
Score distributions / simulations
Winner and O/U probabilities

Trading and validation engine
Implied probability and no-vig markets
Expected value and price comparison
Hedged arbitrage stake allocation
Backtesting and exposure controls

Prepared as a methodological guide. All prices, team statistics, predictions and payoffs in worked examples are hypothetical; this report is not a live betting recommendation.

1. Executive summary and strategy architecture

The system seeks to identify pricing discrepancies by estimating team scoring productivity, adjusting it for the opponent, forecasting pace, and translating the projected scores into probabilities. An independent market module compares those probabilities with available odds, while a cross-book module tests for exact, fully covered arbitrages.



Research priority: establish whether opponent-adjusted PPP produces reliable *out-of-sample* score probabilities before adding complex features or committing capital. A predictive edge must survive transaction costs, price movement and model uncertainty.

2. Offensive and defensive points per play

2.1 Core definitions

Points per play (PPP) normalizes scoring for offensive volume. It differentiates a team that scores frequently because it has many possessions from a team that is genuinely efficient on each offensive play.

$$O_i = \frac{\text{Offensive points scored by team } i}{\text{Offensive plays by team } i},$$
$$D_i = \frac{\text{Offensive points allowed by team } i}{\text{Opponent offensive plays against } i}.$$

A larger O_i indicates a stronger offense; a *smaller* D_i indicates a stronger defense. Use consistent play definitions across games, excluding non-offensive scoring from offensive PPP. Attribute defensive and special-teams touchdowns separately in the eventual total-score model.

Hypothetical team	Offensive PPP	Defensive PPP allowed
Team A	0.480	0.380
Team B	0.360	0.440
League average	0.400	0.400

Team A's observed offense and defense appear stronger, but those numbers are not yet adjusted for opponent strength or the reliability of the sample.

2.2 A simple matchup adjustment

Let $\bar{r}$ denote the relevant league-average offensive PPP. One interpretable baseline estimates team i's scoring efficiency against opponent j as:

$$\boxed{r_{i,j} = \bar{r}\left(\frac{O_i}{\bar{r}}\right)\left(\frac{D_j}{\bar{r}}\right) = \frac{O_i D_j}{\bar{r}}.}$$

The multiplicative assumption means offensive strength and defensive vulnerability interact proportionally. Applying it to the hypothetical teams gives:

$$r_{A,B} = \frac{0.480(0.440)}{0.400} = 0.528, \qquad r_{B,A} = \frac{0.360(0.380)}{0.400} = 0.342.$$

This is a transparent baseline, *not* by itself a complete schedule adjustment: input O_i and D_j are still influenced by which teams each side has faced.

2.3 Opponent-adjusted production model

For a stronger implementation, estimate offense and defense ratings jointly from historical game observations:

$$\log r_{i,j,t} = \alpha + o_{i,t} + d_{j,t} + \beta^{\mathsf{T}} X_t.$$

Here $o_{i,t}$ is team i's latent offensive strength, $d_{j,t}$ is opponent j's defensive vulnerability, and X_t includes game-level information (such as venue, weather, injuries and rest). Constrain or center team effects for identifiability. Apply regularization and time decay to reduce overfitting and downweight stale games.

> **College football consideration:** large strength-of-schedule differences make opponent adjustment especially consequential. Shrink sparse or early-season estimates toward sensible conference- or team-level priors; account for transfers and coaching changes rather than treating last year's roster as unchanged.

3. From efficiency to expected final score

3.1 Projecting the number of plays

PPP alone cannot forecast scoring without a model for pace. At the team level:

$$\hat{P}_i = \hat{N}_i \, \hat{L}_i,$$

where $\hat{N}_i$ is expected offensive drives and $\hat{L}_i$ is expected plays per drive. Model the two teams' game pace jointly because their possessions share a single game clock. Relevant drivers include play-calling tendency, pass rate, stoppages, run/pass mix and anticipated game script.

3.2 Expected points

$$\boxed{\mathbb{E}[S_i] = \hat{P}_i \, r_{i,j} + \delta_i,}$$

where δ_i accounts for expected scoring contributions not already captured in offensive PPP (for example, defensive or special-teams scores). Avoid counting an effect both in $r_{i,j}$ and δ_i.

Hypothetical estimate	Team A	Team B
Expected offensive plays	62	61
Opponent-adjusted PPP (baseline)	0.528	0.342
Projected offensive points	32.736	20.862
Additional scoring adjustment δ_i	0	0

Thus the illustrative forecast is:

$$\boxed{\mu_A = 32.736, \quad \mu_B = 20.862, \quad \mu_T = 53.598, \quad \mu_M = 11.874.}$$

Here $T = S_A + S_B$ is game total and $M = S_A - S_B$ is Team A's scoring margin. Expectations do not tell us how often a team wins or how frequently a market line is exceeded; that requires a score distribution.

4. Outcome probabilities and score distributions

4.1 Baseline normal approximations

As a tractable first pass, assume the scoring margin M and total T are approximately normal:

$$M \sim \mathcal{N}(\mu_M, \sigma_M^2),$$
$$T \sim \mathcal{N}(\mu_T, \sigma_T^2).$$

Then, ignoring discrete tie mass in the baseline approximation,

$$P(\text{Team A wins}) \approx \Phi\left(\frac{\mu_M}{\sigma_M}\right),$$
$$P(\text{Over } L) \approx 1 - \Phi\left(\frac{L - \mu_T}{\sigma_T}\right).$$

Φ is the standard-normal cumulative distribution function; L is the sportsbook's total line. The variances should be estimated from historical out-of-sample residuals, not chosen arbitrarily for live bets. If team scores are correlated,

$$\text{Var}(M) = \text{Var}(S_A) + \text{Var}(S_B) - 2\text{Cov}(S_A, S_B),$$
$$\text{Var}(T) = \text{Var}(S_A) + \text{Var}(S_B) + 2\text{Cov}(S_A, S_B).$$

Illustrative probabilities. Taking $\sigma_M = 13$ and $\sigma_T = 13.5$ only for this example yields:

$$P(A \text{ wins}) \approx \Phi(11.874/13) = 81.95\%,$$

$$P(\text{Over } 51.5) \approx 1 - \Phi((51.5 - 53.598)/13.5) = 56.17\%.$$

A real model must handle discrete football scores, pushes, and the NFL's possibility of ties explicitly. For NCAA football, apply the applicable overtime rules and sportsbook settlement terms.

4.2 Preferred approach: game-level Monte Carlo

A more realistic implementation samples shared game states and drives instead of treating two final scores as independent draws. A simulation should include the joint pace, scoring probabilities for possessions, turnovers, field position, special-teams effects, and game-state feedback.

1. Draw game context and initial drive/pace parameters conditional on the matchup.

2. Simulate possessions and their outcomes, updating time, score differential, field position and expected play-calling.
3. Generate full-time scores $(S_A^{(k)}, S_B^{(k)})$ under the relevant league rules.
4. Repeat for $k = 1, \dots, N$; convert simulated frequencies into market probabilities.

$$\widehat{P}(A \text{ wins}) = \frac{1}{N}\sum_{k=1}^{N} \mathbf{1}\{S_A^{(k)} > S_B^{(k)}\},$$
$$\widehat{P}(\text{Over } L) = \frac{1}{N}\sum_{k=1}^{N} \mathbf{1}\{S_A^{(k)} + S_B^{(k)} > L\}.$$

The shared simulations also yield spread-cover probabilities, push probabilities, score quantiles and correlated outcomes such as *Team A wins and Over 51.5*. Check simulation uncertainty, probability calibration and sensitivity to modeling assumptions.

5. Translate probabilities into market prices

5.1 Implied probabilities and expected value

For decimal odds d, the simple break-even probability is $p_{\text{BE}} = 1/d$. American odds a convert as:

$$p_{\text{implied}} = \begin{cases} \dfrac{100}{a+100}, & a > 0, \\ \dfrac{|a|}{|a|+100}, & a < 0. \end{cases}$$

Because sportsbook prices contain a margin (*vig*), the sum of implied probabilities across an exhaustive market usually exceeds 100%. For a rough no-vig comparison, normalize each outcome's implied probability by the total, while recognizing this is one of several possible margin-removal methods.

For a simple two-outcome bet with no push, expected net profit per dollar staked is:

$$\boxed{EV = p(d-1) - (1-p) = pd - 1.}$$

Example: a 56% modeled win probability at American odds -110 corresponds to $d = 1.909091$ and

$$EV = 0.56(0.909091) - 0.44 \approx 6.91\%.$$

This is an expectation across repeated bets *if the model is correctly calibrated*, not a guaranteed return from one ticket. With a push possibility, use

$$EV = P(\text{win})(d-1) - P(\text{loss}).$$

5.2 Pricing discipline

Use edge $= p_{\text{model}} - p_{\text{BE}}$ as a screening diagnostic, but require a sufficiently large margin above model and transaction uncertainty before any positive-EV designation. Compare against the best *executable* line, not stale screenshots or a price no longer available.

6. Cross-sportsbook arbitrage

6.1 The exact condition

A true two-way arbitrage exists if the same fully covered, identically settled market has opposing decimal prices d_1, d_2 such that

$$\boxed{S = \frac{1}{d_1} + \frac{1}{d_2} < 1.}$$

For a stake budget B, allocate the two wagers to equalize gross payoff:

$$\boxed{B_i = B\,\frac{1/d_i}{S}}, \qquad \boxed{R_{\text{arb}} = \frac{1}{S} - 1}.$$

The model's score prediction is *not necessary* to prove this price identity. Its role is finding markets worth inspecting; only the full, compatible price set establishes an arbitrage.

6.2 Worked Over/Under example

Outcome	Book	Decimal odds	Stake on $100
Over 51.5	Book A	2.10	$49.40
Under 51.5	Book B	2.05	$50.60

$$S = \frac{1}{2.10} + \frac{1}{2.05} = 0.963995,$$
$$\text{Equalized gross payout} = \frac{\$100}{S} \approx \$103.73,$$
$$\text{Theoretical net profit} \approx \$3.73, \qquad R_{\text{arb}} = 3.7349\%.$$

> **Execution is the limiting factor.** Verify that both prices can be accepted for the required stakes, that the total and settlement rules match (including overtime, pushes, voids and corrections), and that fees or other constraints do not erase the payoff. One unfilled leg can convert a hedged trade into an exposed wager.

Important separation: model-based value betting carries outcome risk; a correctly constructed arbitrage removes modeled outcome risk but retains operational, counterparty and rule-mismatch risk. Track performance separately.

7. NFL and college football require separate calibration

Dimension	NFL	College football
Team strengths	More concentrated competition	Wide differences across teams and divisions
Schedule effect	Relevant, with more connected comparisons	Critical, especially across conferences
Roster continuity	Monitor quarterback and position-group availability	Transfers, graduation and coaching turnover important
Early season	Use prior-year and offseason information	Stronger prior shrinkage and uncertainty needed
Scoring and pace	Calibrate to NFL-specific clock/overtime rules	Calibrate to college clock/overtime rules
Forecast error	Empirically estimated by matchup type	Expect heterogeneous uncertainty for mismatches

For NFL, emphasize recent efficiency, injury-adjusted lineup quality, weather, rest, offensive-line matchups, and context. For college football, emphasize schedule-adjusted PPP, changing personnel and potentially high dispersion between opponents.

A hierarchical/time-updating prior can reduce early-season overreaction:

$$\widehat{O}_{i,t} = w_t O_{i,\text{observed}} + (1 - w_t) O_{i,\text{prior}}, \qquad 0 \le w_t \le 1.$$

Increase w_t as informative current-season data accumulates; revise the prior for material roster/coaching changes. Learn weights from historical validation rather than selecting them solely by intuition.

8. Data and implementation architecture

Module	Primary responsibility	Implementation
Data pipeline	Time-stamped play-by-play, game context, injury and odds snapshots	pandas, NumPy, SQL
Efficiency engine	Team ratings, opponent adjustment, pace	statsmodels, scikit-learn
Game simulator	Joint score paths and event probabilities	NumPy, SciPy
Pricing engine	Fair odds, vig adjustment, EV, arb math	Python, market rules
Dashboard	Forecasts, quotes, alerts and exposure	Plotly, Dash

Useful data starting points: nflfastR / nflverse for NFL play-by-play, and CollegeFootballData (CFBD) for college game and team data. Verify access terms, coverage, definitions, update lags and historical point-in-time availability before using them in production.

Store observations and market quotes with *timestamps*, source identifiers and consistent event IDs. A historical betting decision must be replayed against the exact information and odds actually observable before kickoff.

9. Backtesting, exposure and validation

9.1 Evaluation metrics

Metric	What it evaluates
Brier score and log loss	Probability forecast quality, including confidence penalties
Calibration curve	Whether nominal p predictions occur at approximately frequency p
Total-points MAE	Magnitude of errors in expected game totals
Winner/spread accuracy	Directional skill and ability to identify line mispricing
Closing-line value (CLV)	Executed price relative to later market consensus; not proof of profit
Realized ROI / drawdown	Economic performance and downside risk after costs

Use rolling chronological train/validation/test windows. Do not let future injuries, roster information, post-game statistics or closing odds leak into a pregame forecast. Segment results by league, season, edge bucket, favorite/underdog, and uncertainty regime. Compare progressively: raw PPP $\rightarrow$ adjusted PPP $\rightarrow$ full context $\rightarrow$ joint simulation $\rightarrow$ market-informed pricing.

9.2 Walk-forward testing protocol

Each week or game round, freeze the forecast as of a specified pre-kickoff timestamp. Refit using only prior observations, score the next set of games, and archive the then-executable lines. Benchmark against a no-vig market forecast, a simple unadjusted-PPP model and an opponent-adjusted baseline. Report confidence intervals for apparent edge, examine high-uncertainty matchups separately, and run sensitivity tests for key ratings and assumed pace.

9.3 Capital and operational risk

For simple binary value bets, the unconstrained Kelly fraction is

$$\boxed{f^* = \frac{bp - (1-p)}{b}}, \qquad b = d - 1.$$

Because estimated probabilities are uncertain, fractional Kelly and hard event/portfolio caps are often more prudent than unconstrained sizing. Aggregate exposure across moneyline, spread and total positions; their payoffs may be strongly correlated. For arbitrage, independently validate all legs, available limits, matching market definitions and realized execution prices.

10. Formula sheet and build sequence

Question	Core mathematical relationship
How efficiently will team i score?	$r_{i,j} = O_i D_j / \bar{r}$ (baseline; adjust ratings for schedule)
How many points?	$\mathbb{E}[S_i] = \hat{P}_i r_{i,j} + \delta_i$
Who wins?	$P(S_A > S_B)$ from the joint score distribution
Will the total clear L?	$P(S_A + S_B > L)$ from the same distribution
Does a bet have value?	$EV = p(d-1) - (1-p)$ when no push is possible
Is there a full-cover arb?	$\sum_i d_i^{-1} < 1$ for mutually exclusive/exhaustive identical-settlement outcomes
What is the arb allocation?	$B_i = B(d_i^{-1})/(\sum_j d_j^{-1})$

1. **Baseline:** clean play-by-play data; fit rolling, opponent-adjusted PPP ratings and a pace model.
2. **Forecast:** produce score means, margin and total distributions; calibrate on untouched future periods.
3. **Simulation:** model correlated possessions and calculate coherent winner, spread, O/U and joint probabilities.
4. **Market layer:** compare timestamped executable odds; keep positive-EV signals separate from verified arbitrage identities.
5. **Control layer:** apply model-uncertainty thresholds, stake caps, matching-rule checks and post-trade performance monitoring.

> **Bottom line:** The research edge comes from estimating *probabilities*, not just predicting winners. The trading edge comes from comparing those calibrated probabilities with executable prices. Guaranteed mathematical arbitrage is a separate price-and-settlement condition, not the automatic result of an accurate model.

Notation at a glance

Symbol	Meaning	Symbol	Meaning
O_i, D_i	Raw offensive / defensive PPP	$r_{i,j}$	Expected matchup PPP
$\hat{P}_i$	Forecast offensive plays	S_i	Final points for team i
T, M	Game total / A-minus-B margin	L	Market total line
p, d	Model win probability / decimal odds	B_i, S	Hedge stake / sum of inverse prices

Methodological note: All worked team statistics, odds and estimated standard deviations are illustrative and are not claims about current NFL or NCAA games. This document is educational, not individualized financial or wagering advice.